

Activity- 1

AIM:

To assemble the components of a given electrical circuit.

APPARATUS AND MATERIALS

Apparatus: A voltmeter and an ammeter of appropriate range, a battery, a rheostat, a one-way key

Materials: An unknown resistance or resistance coil, connecting wires, a piece of sandpaper

PROCEDURE:

1. Connect the components as shown in the diagram.
2. After closing the key K, check that the voltmeter and ammeter show deflections on the right-hand side.
3. Check the continuity of the assembled circuit using a multimeter.

RESULT:

The components of the electrical circuit were assembled

Precautions

- i. The connections should be neat, clean, and tight.
- ii. All precautions regarding the polarity should be observed
- iii. Zero correction, if any, must be applied.

Sources of Error

- i. The instrument screws may be loose.
- ii. Thick connecting wires may not be available.
- iii. A rheostat may have high resistance.

Diagram and observation table on the left side of the page

Diagram:

